

HOMEWORK 9

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 3.2 - POWER SERIES AND TAYLOR'S THEOREM

1. Compute the Taylor series of the following:

(a) $\frac{\sin z}{z}, z_0 = 1$

Note that computing the Taylor series for $\frac{\sin z}{z}$ at $z_0 = 1$ is equivalent to computing the Taylor series of $\frac{\sin(1-z)}{1-z}$ at $z = 0$. Note that $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$. So, $\frac{\sin(1-z)}{1-z} = \frac{e^{i(1-z)} - e^{-i(1-z)}}{2i(1-z)}$. Expanding, this gives:

$$\begin{aligned} \frac{\sin(1-z)}{1-z} &= \frac{e^{i(1-z)} - e^{-i(1-z)}}{2i(1-z)} \\ &= \frac{1}{2i(1-z)} \left[\sum_{k=0}^{\infty} \left(\frac{(i)^k (1-z)^k}{k!} \right) - \sum_{k=0}^{\infty} \left(\frac{(-i)^k (1-z)^k}{k!} \right) \right] \\ &= \sum_{k=0}^{\infty} \left(\frac{((i)^k - (-i)^k)(1-z)^{k-1}}{2i(k!)} \right) \end{aligned}$$

Since $(i)^0 - (-i)^0 = 0$, we can re-write this as

$$\frac{\sin(1-z)}{1-z} = \sum_{k=0}^{\infty} \frac{(i)^{k+1} - (-i)^{k+1}}{2i(k+1)!} (1-z)^k.$$

(b) $z^2 e^z, z_0 = 0$

Note that the Taylor series of e^z at $z_0 = 0$ is given by $e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!}$. So, we have $z^2 e^z = \sum_{k=0}^{\infty} \frac{z^{k+2}}{k!}$. That is, writing this series in the form $\sum_{k=0}^{\infty} a_k (z - z_0)^k$, we have $a_0 = a_1 = 0$ and $a_k = \frac{1}{(k-2)!}$ for $k \geq 2$.

(c) $e^z \sin z, z_0 = 0$

Note that $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$. So, $e^z \sin z = \frac{e^{(1+i)z} - e^{(1-i)z}}{2i}$. Expanding, this gives:

$$\begin{aligned} e^z \sin z &= \frac{e^{(1+i)z} - e^{(1-i)z}}{2i} \\ &= \frac{1}{2i} \left[\sum_{k=0}^{\infty} \left(\frac{(1+i)^k z^k}{k!} \right) - \sum_{k=0}^{\infty} \left(\frac{(1-i)^k z^k}{k!} \right) \right] \\ &= \frac{1}{2i} \sum_{k=0}^{\infty} \frac{[(1+i)^k - (1-i)^k] z^k}{k!}. \end{aligned}$$

That is, writing this series in the form $\sum_{k=0}^{\infty} a_k (z - z_0)^k$, we have $a_k = \frac{(1+i)^k - (1-i)^k}{2i(k!)}$.

2. Compute the first few terms in the Taylor expansion of $\sqrt{z^2 - 1}$ around 0.

The Taylor expansion of an analytic function at z_0 is of the form $f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k$. Setting

$f(z) = \sqrt{z^2 - 1}$, we have

$$\begin{aligned} f(0) &= \sqrt{z^2 - 1} \Big|_{z=0} = i \\ f^{(1)}(0) &= \frac{z}{\sqrt{z^2 - 1}} \Big|_{z=0} = 0 \\ f^{(2)}(0) &= \frac{-1}{(z^2 - 1)^{3/2}} \Big|_{z=0} = -i \\ f^{(3)}(0) &= \frac{3z}{(z^2 - 1)^{5/2}} \Big|_{z=0} = 0 \\ f^{(4)}(0) &= \frac{3(4z^2 + 1)}{(z^2 - 1)^{7/2}} \Big|_{z=0} = -3i \\ f^{(5)}(0) &= \frac{15z(4z^2 + 3)}{(z^2 - 1)^{9/2}} \Big|_{z=0} = 0 \\ f^{(6)}(0) &= \frac{45(8z^4 + 12z^2 + 1)}{(z^2 - 1)^{11/2}} \Big|_{z=0} = -45i \end{aligned}$$

So, the first few terms of the Taylor expansion of $\sqrt{z^2 - 1}$ around 0 are given by

$$f(z) \approx \frac{i}{0!} - \frac{i}{2!} z^2 - \frac{3i}{4!} z^4 - \frac{45i}{6!} z^6 = i - \frac{i}{2} z^2 - \frac{i}{8} z^4 - \frac{i}{16} z^6.$$

3. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converge for $|z| < R$. If $0 < r < R$, show that $f(z) = \sum_{n=0}^{\infty} a_n r^n e^{in\theta}$, where $z = re^{i\theta}$ and

$$a_n = \frac{1}{2\pi r^n} \int_0^{2\pi} f(re^{i\theta}) e^{-in\theta} d\theta.$$

Also show that

$$\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.$$

Hint: Use the Cauchy Integral Formula for a_n and expand $f\bar{f}$ in a series. Then, integrate term by term.

Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ converge for $|z| < R$ and suppose $0 < r < R$. Note that

$$f(re^{i\theta}) = \sum_{n=0}^{\infty} a_n (re^{i\theta})^n = \sum_{n=0}^{\infty} a_n r^n e^{in\theta}.$$

Now, observe that

$$\begin{aligned} \frac{1}{2\pi r^n} \int_0^{2\pi} f(re^{i\theta}) e^{-in\theta} d\theta &= \frac{1}{2\pi} \int_0^{2\pi} \frac{f(re^{i\theta})}{r^n e^{in\theta}} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{f(re^{i\theta})}{(re^{i\theta})^n} d\theta \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(re^{i\theta})}{(re^{i\theta})^{n+1}} i re^{i\theta} d\theta \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(re^{i\theta})}{(re^{i\theta})^{n+1}} \frac{d}{d\theta}(e^{i\theta}) d\theta \\ &= \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z^{n+1}} dz \quad (\text{Letting } \gamma(\theta) = re^{i\theta}). \end{aligned}$$

Via the Cauchy integral formula, this is just $\frac{f^{(n)}(0)}{n!}$. So, by the uniqueness of the Taylor series,

$$a_n = \frac{f^{(n)}(0)}{n!} = \frac{1}{2\pi r^n} \int_0^{2\pi} f(re^{i\theta}) e^{-in\theta} d\theta.$$

Now, we show that

$$\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.$$

To begin, recall that $\overline{e^{i\theta}} = e^{-i\theta}$. So, if $f(z) = \sum_{n=0}^{\infty} a_n r^n e^{in\theta}$, then $\bar{f}(z) = \sum_{n=0}^{\infty} \bar{a}_n r^n e^{-in\theta}$. So, since $f\bar{f} = |f|^2$, we have:

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta &= \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) \bar{f}(re^{i\theta}) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left(\sum_{n=0}^{\infty} a_n r^n e^{in\theta} \right) \left(\sum_{n=0}^{\infty} \bar{a}_n r^n e^{-in\theta} \right) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \left(\sum_{k=0}^{\infty} \sum_{n=0}^{\infty} a_k r^k e^{ik\theta} \bar{a}_n r^n e^{-in\theta} \right) d\theta \end{aligned}$$

Note that when $k = n$, $a_n r^n e^{in\theta} \overline{a_n} r^n e^{in\theta} = |a_n|^2 r^{2n}$. When $k \neq n$, then we have $a_k r^k e^{ik\theta} \overline{a_n} r^n e^{in\theta} = a_k \overline{a_n} r^{n+k} e^{i(k-n)\theta}$. Since $\int_0^{2\pi} a_k \overline{a_n} r^{n+k} e^{i(k-n)\theta} d\theta = a_k \overline{a_n} r^{n+k} \int_0^{2\pi} e^{i(k-n)\theta} d\theta = 0$ by Cauchy's Theorem, we produce

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} \left(\sum_{k=0}^{\infty} \sum_{n=0}^{\infty} a_k r^k e^{ik\theta} \overline{a_n} r^n e^{-in\theta} \right) d\theta &= \frac{1}{2\pi} \left(\sum_{k=0}^{\infty} \sum_{n=0}^{\infty} \int_0^{2\pi} a_k r^k e^{ik\theta} \overline{a_n} r^n e^{-in\theta} d\theta \right) \\ &= \frac{1}{2\pi} \sum_{n=0}^{\infty} \int_0^{2\pi} |a_n|^2 r^{2n} d\theta \\ &= \frac{1}{2\pi} \left(\sum_{n=0}^{\infty} 2\pi |a_n|^2 r^{2n} \right) \\ &= \sum_{n=0}^{\infty} |a_n|^2 r^{2n} \end{aligned}$$

Therefore, we have shown that

$$\frac{1}{2\pi} \int_0^{2\pi} \left| f(re^{i\theta}) \right|^2 d\theta = \sum_{n=0}^{\infty} |a_n|^2 r^{2n}.$$

4. Find a function such that $f(0) = 1$ and $f'(x) = x + 2f(x)$ for all x .

Let us suppose that f is the restriction to the real axis of a function that is analytic in $\mathbb{C}$ (i.e. of an entire function). Therefore, it will have a Taylor series centered at zero which converges on $\mathbb{C}$.

That is, $f(z) = \sum_{k=0}^{\infty} a_k z^k$ for all $z \in \mathbb{C}$ and $f'(z) = \sum_{k=1}^{\infty} k a_k z^{k-1}$. So, note that

$$\sum_{k=1}^{\infty} k a_k z^{k-1} = z + 2 \sum_{k=0}^{\infty} a_k z^k.$$

That is,

$$\sum_{k=0}^{\infty} (k+1) a_{k+1} z^k = 2a_0 + (1+2a_1)z + \sum_{k=2}^{\infty} (2a_k) z^k.$$

Or, more simply,

$$(2a_0 - a_1) + (1 + 2a_1 - 2a_2)z + \sum_{k=2}^{\infty} (2a_k - (k+1)a_{k+1})z^k = 0.$$

Since $f(0) = 1$ and $f(z) = \sum_{k=0}^{\infty} a_k z^k$, it must be the case that $a_0 = 1$. Since $2a_0 - a_1 = 0$, we then have $a_1 = 2$. Similarly, $1 + 2a_1 - 2a_2 = 0$, so $a_2 = \frac{5}{2}$. For $k \geq 2$, we have $2a_k - (k+1)a_{k+1} = 0$, so $a_{k+1} = \frac{2}{k+1}a_k$. That is, $a_0 = 1$, $a_1 = 2$, $a_2 = \frac{5}{2}$, and

$$\begin{aligned} a_3 &= \left(\frac{2}{3}\right)a_2 \\ a_4 &= \left(\frac{2}{4}\right)a_3 = \left(\frac{2}{4}\right)\left(\frac{2}{3}\right)a_2 \\ a_5 &= \left(\frac{2}{5}\right)a_4 = \left(\frac{2}{5}\right)\left(\frac{2}{4}\right)a_3 = \left(\frac{2}{5}\right)\left(\frac{2}{4}\right)\left(\frac{2}{3}\right)a_2 \\ &\vdots \end{aligned}$$

More generally, for $k \geq 3$, we have $a_k = \frac{2^{k-2}}{(k!)/2} a_2 = 5 \frac{2^{k-2}}{k!}$. Notice that this also gives $a_2 = 5 \frac{2^{2-2}}{2!} = \frac{5}{2}$, so we have

$$\begin{aligned} f(z) &= 1 + 2z + 5 \sum_{k=2}^{\infty} \frac{2^{k-2}}{k!} z^k \\ &= 1 + 2z + \frac{5}{4} \sum_{k=2}^{\infty} \frac{(2z)^k}{k!} \\ &= 1 + 2z + \frac{5}{4} (e^{2z} - 1 - 2z) \\ &= \frac{1}{4} (5e^{2z} - 2z - 1). \end{aligned}$$

Notice that we may check our work: $f(0) = \frac{1}{4} (5e^{2(0)} - 2(0) - 1) = \frac{1}{4} (5 - 1) = 1$ and since

$$f'(z) = \frac{10}{4} e^{2z} - \frac{2}{4} = \frac{5}{2} e^{2z} - \frac{1}{2},$$

we have

$$z + 2f(z) = z + \frac{2}{4} (5e^{2z} - 2z - 1) = z + \frac{5}{2} e^{2z} - z - \frac{1}{2} = f'(z).$$

SECTION 3.3 - LAURENT SERIES AND THE CLASSIFICATION OF SINGULARITIES

5. Find the Laurent series expansion of $\frac{z}{z+1}$ around $z_0 = 0$ valid in the region $1 < |z| < \infty$.

For $|z| < 1$, note that $\frac{1}{1-z} = 1 + z + z^2 + \dots$. So, for $1 < |z| < \infty$, we have $0 < \left|\frac{1}{z}\right| < 1$ and it follows that

$$\frac{z}{z+1} = \left(\frac{1/z}{1/z}\right) \left(\frac{z}{z+1}\right) = \frac{1}{1 - \left(-\frac{1}{z}\right)} = 1 + \left(-\frac{1}{z}\right) + \left(-\frac{1}{z}\right)^2 + \left(-\frac{1}{z}\right)^3 + \dots$$

That is, the Laurent expansion of $\frac{z}{z+1}$ around $z_0 = 0$ that is valid in the region $1 < |z| < \infty$ is given by

$$\frac{z}{z+1} = \sum_{k=0}^{\infty} \left(-\frac{1}{z}\right)^k = 1 - \frac{1}{z} + \frac{1}{z^2} - \frac{1}{z^3} + \dots$$

6. Prove, using the Taylor series, the following complex version of l'Hopital's rule:

Let $f(z)$ and $g(z)$ be analytic, both having zeros of order k at z_0 . then, $f(z)/g(z)$ has a removable singularity and

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f^{(k)}(z_0)}{g^{(k)}(z_0)}.$$

Let $f(z)$ and $g(z)$ be analytic, both having zeros of order k at z_0 . By Proposition 3.2.9, there exists analytic functions ϕ, ψ such that $\phi(z_0) \neq 0$, $\psi(z_0) \neq 0$, and such that $f(z) = (z - z_0)^k \phi(z)$, $g(z) = (z - z_0)^k \psi(z)$ for all z in a disk of nonzero radius about z_0 . So,

$$\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{(z - z_0)^k \phi(z)}{(z - z_0)^k \psi(z)} = \frac{\phi(z_0)}{\psi(z_0)}.$$

Now, by Taylor's Theorem, observe that the analyticity of f and g implies that there are some constants $r_1, r_2 > 0$ such that the Taylor series for $f(z)$ and $g(z)$ converge on $D(z_0, r_1)$ and $D(z_0, r_2)$, respectively. Taking r to be the lesser of these two values, we have

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n \quad \text{and} \quad g(z) = \sum_{n=0}^{\infty} \frac{g^{(n)}(z_0)}{n!} (z - z_0)^n$$

on $D(z_0, r)$. Since each $f(z)$ and $g(z)$ have a zero of order k at z_0 , it follows that $f^{(n)}(z_0) = g^{(n)}(z_0) = 0$ for $n = 0, 1, \dots, k$ and that $f^{(k)}(z_0) \neq 0$, $g^{(k)}(z_0) \neq 0$. So, we have

$$f(z) = \sum_{n=k}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n \quad \text{and} \quad g(z) = \sum_{n=k}^{\infty} \frac{g^{(n)}(z_0)}{n!} (z - z_0)^n$$

on $D(z_0, r)$. Accordingly,

$$\sum_{n=k}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n = (z - z_0)^k \phi(z) \quad \text{and} \quad \sum_{n=k}^{\infty} \frac{g^{(n)}(z_0)}{n!} (z - z_0)^n = (z - z_0)^k \psi(z)$$

on some disk of nonzero radius centered at z_0 and contained in $D(z_0, r)$, so

$$\phi(z) = \sum_{n=0}^{\infty} \frac{f^{(n+k)}(z_0)}{n!} (z - z_0)^n \quad \text{and} \quad \psi(z) = \sum_{n=0}^{\infty} \frac{g^{(n+k)}(z_0)}{n!} (z - z_0)^n$$

on this disk. But, by the analyticity of ϕ and ψ together with the uniqueness of power series (i.e., Theorem 3.2.5), these must be the Taylor series of ϕ and ψ ! So,

$$\phi(z_0) = f^{(k)}(z_0) \quad \text{and} \quad \psi(z_0) = g^{(k)}(z_0).$$

Combining with our observation that $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{\phi(z_0)}{\psi(z_0)}$, this produces our desired result.